AI-Based Loan Default Intelligence System

Comprehensive Technical & Architecture Report

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1. Executive Summary

The AI-Based Loan Default Intelligence System is a modern, full-stack predictive platform designed for the financial sector. Unlike traditional "black box" machine learning models that simply output a binary "Approve/Reject," this system is built on a "Three-Pillar" architecture:

1. **Prediction:** Accurate risk assessment using state-of-the-art ensemble models (XGBoost).
2. **Interpretability:** Mathematical unpacking of risk drivers via SHAP (SHapley Additive exPlanations).
3. **Actionability:** Translation of complex data arrays into natural language summaries using Generative AI (Google Gemini 2.5 Flash).

This ensures compliance, builds trust with loan officers, and drastically reduces manual application review times.

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2. System Architecture

The ecosystem relies on an automated, horizontally scalable structure. Data flows from raw CSV intake through a robust ML pipeline, culminating in an interactive Streamlit UI powered by a FastAPI backend.

```
graph TD
    %% Define Nodes
    RawData[(Raw Loan Data\nCSV/DB)]
    Preprocess[Data Preprocessing & \nFeature Engineering Pipeline]
    SMOTE[SMOTE\nClass Balancing]
    ModelMain{XGBoost Classifier}

    SHAP[SHAP Explainer\n(Interpretability Layer)]
    LLM[GenAI Module\n(Google Gemini 2.5)]

    FastAPI[FastAPI Backend\nMicroservice]
    Streamlit[Streamlit UI\n(Loan Officer Dashboard)]

    %% Define Flow
    RawData --> Preprocess
    Preprocess --> SMOTE
    SMOTE --> ModelMain

    ModelMain -- Pred: Default Probability --> SHAP
    Preprocess -- Test Applicant Profile --> SHAP

    SHAP -- Top Risk/Mitigating Factors --> LLM
    LLM -- Natural Language Narrative --> FastAPI
```

```
SHAP -- Feature Importance Arrays --> FastAPI
ModelMain -- Risk Score --> FastAPI

FastAPI <--> Streamlit

%% Styling
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classDef process fill:#bbf,stroke:#333,stroke-width:2px;
classDef decision fill:#bfb,stroke:#333,stroke-width:2px;
classDef ui fill:#fbb,stroke:#333,stroke-width:2px;

class RawData database;
class Preprocess,SMOTE process;
class ModelMain,SHAP,LLM decision;
class FastAPI,Streamlit ui;
```

Layer Breakdown

A. Data Layer (`src/data/`, `src/features/`)

- Capable of ingesting LendingClub, Home Credit, or generating realistic 10k-row synthetic data.
- **Preprocessing:** Handles missing continuous variables via median imputation to resist outliers, and categorical variables via mode imputation. It uses `LabelEncoder` for string categories and scales all features using `StandardScaler`.
- **Class Balancing:** Utilizes SMOTE (Synthetic Minority Over-sampling Technique) strictly on the training set to prevent model bias toward the majority "paid off" class.
- **Engineering Highlights:**
  - *Debt-to-Income (DTI) Ratio*
  - *Credit Utilization Ratio*
  - *Loan-to-Income Ratio*

B. Inference Layer (`src/models/`, `src/api/`)

- Uses **XGBoost** for gradient-boosted decision trees. XGBoost was chosen over Logistic Regression for its superior handling of non-linear financial interactions.
- Integrates **SHAP** (`TreeExplainer`) to dissect the prediction. It isolates the top 3 factors pulling the risk score up, and the top factors pulling it down.
- Implements a **Generative AI wrapper** (`DefaultExplainerLLM`). It passes the SHAP array into a constructed prompt directed at Google Gemini, mandating a professional, 3-sentence risk summary for human consumption.

C. Application Layer (`api/`, `app/`)

- Decoupled **FastAPI** interface exposes the models as a REST API (`/analyze_borrower`).
- **Streamlit** dashboard consumes the backend predictions and paints a responsive UI. It visualizes the SHAP values using custom Matplotlib horizontal bar charts (red for risk, green for safety).

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3. Key Differentiators for Hackathon Pitch

When presenting on slides, emphasize these three points to stand out against standard ML projects:

1. **Compliance Through Explainability:** Standard neural networks struggle with credit regulations (like the Equal Credit Opportunity Act) because they cannot explain their rejections. By deploying SHAP, we ensure every decision is mathematically justified.
2. **GenAI as a Translator:** We don't expect a loan officer to read log-loss coefficients. Gemini acts as an autonomous financial analyst bridging the gap between Data Science and Business Operations.
3. **Designed for Deployment:** We didn't leave the code in a Jupyter Notebook. The inclusion of Docker and a FastAPI REST layer proves the system is enterprise-ready and scalable.

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4. Evaluation Metrics (Sample Run)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.820	0.540	0.814	0.649	0.893
Random Forest	0.840	0.590	0.711	0.645	0.869
XGBoost	0.864	0.687	0.613	0.648	0.863

Note: Models evaluated on 20% holdout test set with SMOTE applied to training distributions.